Carlos Inca Roca

carlos.incaroca@gmail.com | linkedin.com/in/carlos-incaroca | carlos-portfolio-nu.vercel.app

EDUCATION

Brigham Young University Provo, UT

MS Mechanical Engineering, in progress

- Advisor: Dr. Steven Charles, Neuromechanics Research Group

BS Mechanical Engineering, Minor in Mathematics April 2026

- GPA 3.78

Carnegie Mellon University, Robotics Institute Pittsburgh, PA

MS Robotic Systems Development (MRSD), admitted Fall 2027

EXPERIENCE

BYU Neuromechanics Research Group Provo, UT

Research Assistant Jan 2025 – Present

- Helped expand the group's upper-limb tremor propagation model from 15 to 50 muscles, producing 3x more simulation data for complete upper-limb analysis.
- Rewrote the simulation code, cutting runtime 85x (from 170 hours to 2 hours).
- Modeled upper-limb dynamics using robotics principles and simulation software to support the group's custom MATLAB model.
- MS research: using the model to study how tremor changes when the limb itself changes, whether through tremor suppression or daily activity.
- Co-authoring a research paper on this work.

Pine and Pepper Provo, UT

Agentic AI Engineering Intern May 2026 – Aug 2026

- Designed and deployed an end-to-end agentic AI system (n8n + Claude LLMs) that autonomously sources, scores, and publishes affiliate deals, replacing the manual workflows of 2 roles and running unattended 3x daily.
- Architected its multi-stage LLM pipeline (rubric-based scoring, auto-classification, deduplication, and brand-safety guardrails), delivering quality-gated output with human-in-the-loop review.
- Ran a systems and financial analysis of the company's affiliate-revenue structure and built a data-driven scoring model that drove $10K+ in annual savings.
- Coordinated across 4 cross-functional teams to map and optimize operational workflows, standardizing hand-offs and eliminating redundant manual steps.

BYU Structural Dynamics Research Group Provo, UT

Research Assistant Jan 2025 – Jan 2026

- Built vibration and damping models in Abaqus for additively manufactured (LPBF) aerospace components using anisotropic and viscoelastic materials, reaching 95.6% damping-ratio prediction accuracy.
- Coordinated research collaborations with partners in Germany and the United Kingdom to extend the analysis to additional LPBF geometries and verify model accuracy.
- Improved part-design procedures for vibration testing before manufacturing, reducing prototyping time.
- Contributed to a short paper, "Applying a Viscoelastic Model to Multiple Unique Additively-Manufactured Trapped-Powder Dampers," submitted to the IMAC conference.

BYU Mathematics Department Provo, UT

Math Lab Teaching Assistant Jan 2023 – Dec 2024

- Trained and mentored 6+ new TAs, improving team efficiency and raising daily student support by 30%.
- Led midterm reviews for 100+ students each semester; supported 60+ students daily in advanced math courses.

OyM Cusco, Peru

Engineering Design Intern May 2021 – Sep 2021

- Designed HVAC systems for COVID-19 hospitals, pools, and hotels using fluid mechanics and heat transfer.
- Designed 14 custom AutoCAD models for mechanical equipment (valves, heaters, filters) and optimized system performance using real-time field data from on-site teams.

ENTREPRENEURSHIP AND LEADERSHIP

SIP Hydration Provo, UT
Co-founder and Lead Mechanical Engineer Sep 2025 – Present

- Co-founded an early-stage startup building a hydration tracker that fits most water bottles; led mechanical design (CAD) and cross-functional product and mechatronics decisions.
- Recruited and led an 11-person team of engineering and business students across 2 sub-teams, coordinating engineering and go-to-market weekly.
- Drove market validation: 50+ customer interviews, 30+ waitlist signups, 2 inbound letters of intent from ads.
- Pitched the venture to investors in a BYU entrepreneurship course (ENT 401).

The Church of Jesus Christ of Latter-day Saints Córdoba, Argentina
Full-time Volunteer Missionary; Zone Leader Jul 2017 – Jul 2019

- Served two years as a full-time volunteer; as zone leader, was responsible for the province of La Rioja and reported to the mission president in Córdoba.

Neighborhood Fathers' Group Peru
Volunteer Group Leader COVID-19 pandemic

- Led a neighborhood group of 30 fathers: organized meetings, checked in on families remotely, and helped distribute donations to families struggling to make ends meet.

ENGINEERING PROJECTS

BYU Mars Rover Team Provo, UT
Mechanical Design Engineer Jan 2025 – May 2026

- Led the mechanical redesign of the rover's robotic arm joints; operated and refined the arm for retrieval and delivery missions requiring precise, real-time control.
- Team placed 4th of 114 teams from 15 countries at the 2025 University Rover Challenge.
- Led the team's transition to a cloud-based CAD platform and mentored new members on integration processes.

Arthropod Robotics Design Project Provo, UT
Team Lead, Mechanical Design Engineer May 2024 – Jun 2025

- Led a team of 4 to design and build a six-legged walking robot, integrating 38 custom components and 18 motors with synchronized gait control on an Arduino R4.
- Implemented inverse kinematics algorithms and dynamic simulations to refine leg trajectories across 5 prototype versions; built a wireless iPad control interface.

Autonomous Robotics Design Project Provo, UT
Hardware and Control Systems Engineer Jan 2025 – Apr 2025

- Designed an autonomous robot for multi-task obstacle courses, integrating sensors for navigation, IR detection, and object manipulation; led hardware design across four prototype iterations.
- Wrote the master control code in C on a PIC microcontroller, unifying subsystem programs and achieving about 50% faster completion times than peer entries.

HONORS AND AWARDS

- Paullin Endowed Fund Engineering Scholarship and Brigham Young Scholarship for academic excellence, BYU
- 3rd Place, 2024 SHPE National Engineering Academic Competition
- 1st Place, University Robotics Competition (30+ entries)
- Selected for the BYU Mars Rover competitive design team

SKILLS AND LANGUAGES

- CAD and simulation: SolidWorks, Onshape, AutoCAD, Ansys, Abaqus FEA
- Programming and embedded systems: MATLAB, Python, C, C++, ROS2, Arduino, PIC
- Engineering and robotics: control systems, dynamic system modeling, robot dynamics and control, mechatronics, mechanical design, machine learning, deep learning
- Languages: Spanish (native), English (fluent), Portuguese